Roll No. | | | | | | | |

End Term Back Examination June 2024

Name of the Program: **B.Tech**

Semester: **III**

Name of the Course: **ECE**

Course Code: **TEC-304**

Paper Name: **Signal & System**

Time: **3 Hour**

Maximum Marks: **100**

Note:

(i) All questions are compulsory
(ii) Answer any two sub questions among a, b and c in each main question
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks

Q1	(10*2= 20 marks)	**CO1**
(a)	Write short notes on the following signals: (i) unit step (ii) unit impulse (iii) unit ramp (iv) signum	
(b)	Examine whether the following signals are periodic or not? If periodic determine the fundamental period. **(i)** $\cos 2t + \sin\sqrt{3}\,t$, **(ii)** $3u(t) + 2\sin 2t$	
(c)	What are the basic operations on signals? Illustrate with an example.	
Q2	(10*2= 20 marks)	**CO2**
(a)	Check whether the following systems are: Static or dynamic, Linear or nonlinear, Causal or non-causal. i) $y(t) = at^2x(t) + btx(t-4)$ ii) $y(n) = a^n u(n)$	
(b)	Define a system. How are systems classified? Define each one of them.	
(c)	Consider a causal LTI system with impulse response $h(t) = u(t)$. Find the output of the system for an input $x(t) = u(t)$.	
Q3	(10*2= 20 marks)	**CO3**
(a)	State and prove the time shifting and frequency shifting properties of Fourier transform.	
(b)	Find the Fourier transform of following signals: $(i) x(t) = u(t)$ $(ii) x(t) = \delta(t)$ $(iii) x(t) = \sin(w_0 t)$	
(c)	Using Fourier transform find the convolution of the signals. $x_1(t) = e^{-2t}u(t)$ and $x_2(t) = e^{-3t}u(t)$.	
Q4	(10*2= 20 marks)	**CO4**
(a)	Find the Laplace transform of the following signals using properties of Laplace transform: (i) $x(t) = u(t-3)$, (ii) $x(t) = e^{-t}u(t)$,	
(b)	State and prove initial and final value theorems based on unilateral Laplace transform.	

(c)	What is ROC? Discuss about ROCs of various classes of signals.	
Q5	(10*2= 20 marks)	CO5
(a)	State and prove the sampling theorem.	
(b)	State and prove the time shifting, time reversal and time expansion properties of Z transforms.	
(c)	Determine the Nyquist rate corresponding the given signal. $x(t) = 1 + \cos 2000\,\pi t + \sin 4000\pi t$	